#9 Intro to

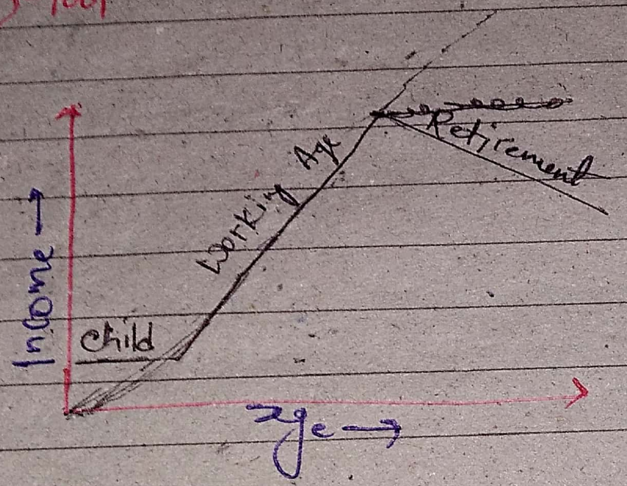
features are the way you represent your knowledge about the world for the Classifier.

These are the way to represent features

Bucketing — Crossing — Hashing — Embedding

Utilities of Tensorflow ~~help~~ Provide to help

Facets — Tool



→ Predicting income of employee using Age.

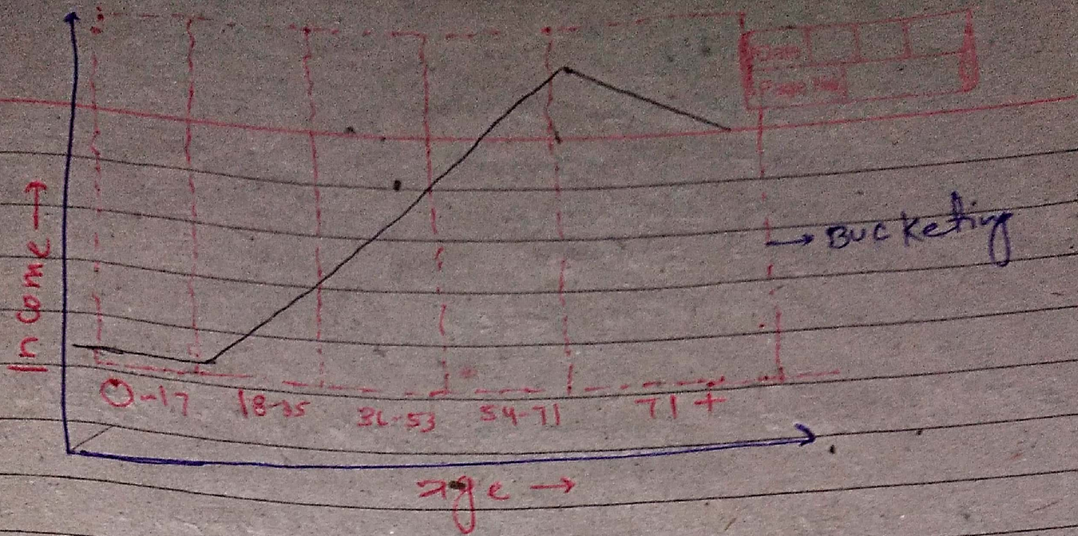
- Its not linear R/L with Income
- A linear Classifier unable to capture this R/L because it learns single weight for each feature.

Bucketing

→ So ~~to~~ to make it easier for classifier one thing we can do, bucket the feature because Bucketing transform several numeric feature into categorical ones based on the range it falls into

→ each of these new feature indicate whether a person's age falls into that range

→ and now a linear model can capture th R/L by learning different weights for each



We can examine it in (Facets) → It is open source visualization tool deployed ~~by~~ to understand and analyze ML datasets. provide Highlevel summary of datasets to quickly grasp overall shape and distribution of their data.

~~use~~ We will use 1 bucketized feature of Facets in this ~~age~~ Income - age data

1) ~~Categorical feature~~

~~Example Education column~~

(1) Categorical feature -

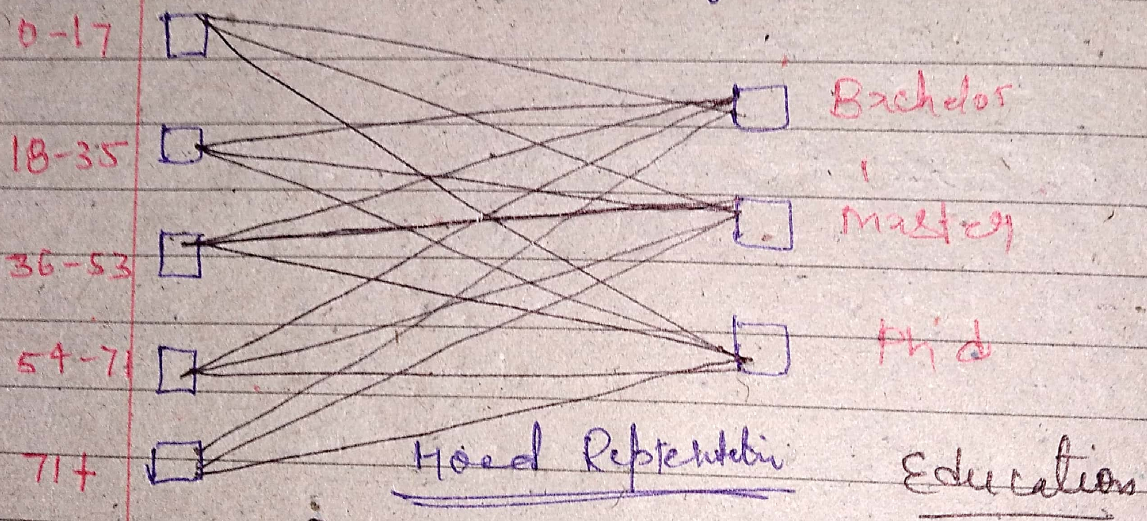
Ex - Education Column
- High School
- Bachelors
- masters
- phd etc.

Small vocabulary - use the raw value

Larger vocabulary - Consider Hashing or Embedding

(2) Feature crossing

It is a way to create a new feature that are combination of existing one. It allow model or linear model to learn non-linear R/L & complex interactions b/w variables. It can generate many possibility quickly & that's why kept under the hashing

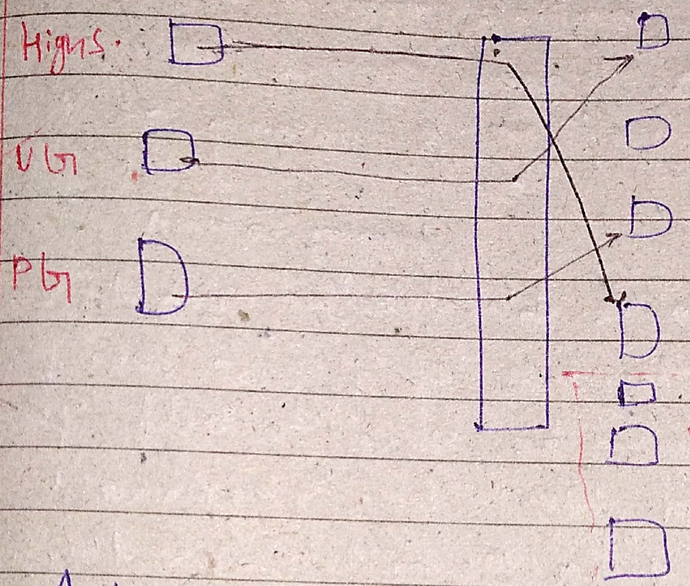


Age

for each cross, we create a new T/F feature

3 Hashing → Can Save memory / Can Save time

- It is a technique that transforms data (like text, number or files) into a fixed-size value using hash fn. This ~~hash~~ hash code act as a unique identifier, making it efficient to store & retrieve in a hash table.
- It is useful for handling a Categorial feature with a large vocabulary or high-Cardinality Categorical data.
- If we don't know the vocab (dataset limit) then we Can use hash function to compute bit automatically.



The downside there could be collision means different items are maped for the same value.

Hash Can also be used to Limit memory usage

A hash Column Can be used to limit the max no. of possibilities, So its Save time

4 Embeddings

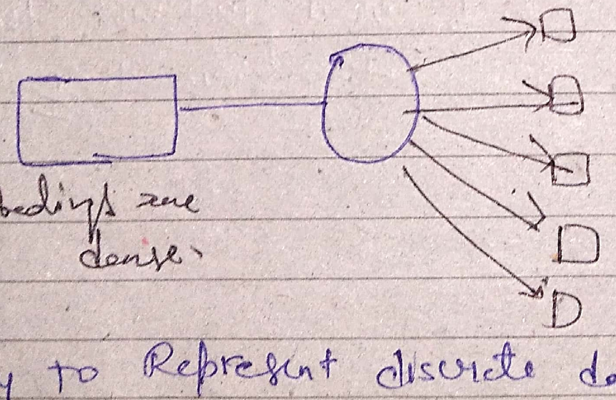
- Powerful way ... Vocabularies.
- Learned Automatically.
- Dense-Vector Vs. One hot (Sparse).

- Powerful way to ~~learn~~ work with Categorical data in a deep learning Setting.
- We can think of an embedding as a vector that represents the meaning of a word
→ We can visualize dataset of word embedding using TF embedding Projector

- It learnt automatically in the process of training a DNN
→ So we need to create embedding column.

if we want to compress the representation so the Classifier learns general Concepts rather than memorizing the meaning of specific words.

for ex:- if we have dataset ^Having column^ called job title. So there are thousands of different jobs & embedding could be used to help your classifier learn words like programmer & software engineer often mean the same things



Embedings are dense.

A way to Represent discrete data (like words, images or other objects) as continuous vector in a lower-dimensional space. Allow ML model to better understand R/L & patterns within the data. Making it easier to perform task like similarity comp., recommendation system